

# 10 Chapter Review

➔ Go to **Mastering Biology** for Assignments, the eText, the Study Area, and Dynamic Study Modules.

## SUMMARY OF KEY CONCEPTS

➔ To review key terms, go to the **Vocabulary Self-Quiz** in the **Mastering Biology** eText or Study Area, or go to [goo.gl/zk9t9t](http://goo.gl/zk9t9t).

### CONCEPT 10.1

#### Photosynthesis feeds the biosphere (p. 188)

- **Photosynthesis** converts the energy of sunlight into chemical energy stored in sugars.
- **Autotrophs** (“producers”) produce their organic molecules from CO<sub>2</sub> and other inorganic raw materials; plants are *photoautotrophs* that use light energy to do this. **Heterotrophs** (consumers, including decomposers) are unable to make their own food and live on compounds produced by others.

? Using the terms *producers*, *consumers*, and *decomposers*, explain how photosynthesis feeds organisms, either directly or indirectly.

### CONCEPT 10.2

#### Photosynthesis converts light energy to the chemical energy of food (pp. 189–192)

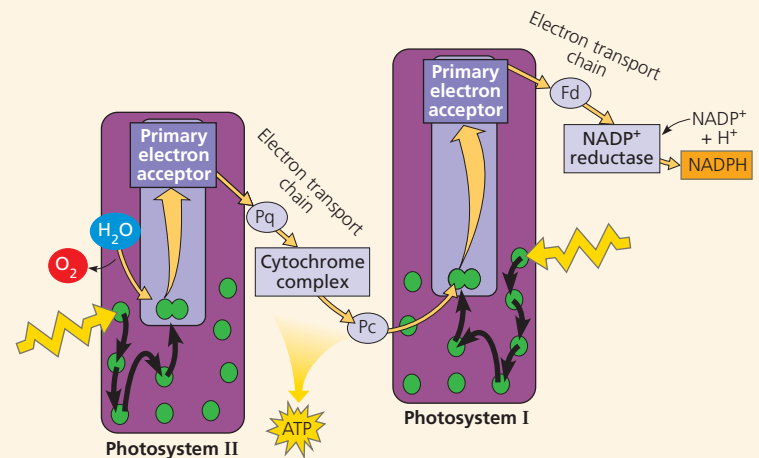
- In plants and other eukaryotic autotrophs, photosynthesis occurs in **chloroplasts**, organelles containing **thylakoids**. Stacks of thylakoids form grana. Photosynthesis is summarized as  $6\text{ CO}_2 + 12\text{ H}_2\text{O} + \text{Light energy} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{ O}_2 + 6\text{ H}_2\text{O}$ . Chloroplasts split water into hydrogen and oxygen, incorporating the electrons of hydrogen into sugar molecules. Photosynthesis is a redox process: H<sub>2</sub>O is oxidized, and CO<sub>2</sub> is reduced. The **light reactions** in the thylakoid membranes split water, releasing O<sub>2</sub>, producing ATP, and forming **NADPH**. The **Calvin cycle** in the **stroma** forms sugar from CO<sub>2</sub>, using ATP for energy and NADPH for reducing power.

? Compare the roles of CO<sub>2</sub> and H<sub>2</sub>O in cellular respiration and photosynthesis.

### CONCEPT 10.3

#### The light reactions convert solar energy to the chemical energy of ATP and NADPH (pp. 192–201)

- Light is a form of electromagnetic energy. The colors we see as **visible light** include those **wavelengths** that drive photosynthesis. A pigment absorbs light of specific wavelengths; **chlorophyll *a*** is the main photosynthetic pigment in plants. Other accessory pigments absorb different wavelengths of light and pass the energy on to chlorophyll *a*.
- A pigment goes from a ground state to an excited state when a **photon** of light boosts one of the pigment’s electrons to a higher-energy orbital. This excited state is unstable. Electrons from isolated pigments tend to fall back to the ground state, giving off heat and/or light.
- A **photosystem** is composed of a **reaction-center complex** surrounded by **light-harvesting complexes** that funnel the energy of photons to the reaction-center complex. When a special pair of reaction-center chlorophyll *a* molecules absorbs energy, one of its electrons is boosted to a higher energy level and transferred to the **primary electron acceptor**. **Photosystem II** contains P680 chlorophyll *a* molecules in the reaction-center complex; **photosystem I** contains P700 molecules.
- **Linear electron flow** during the light reactions uses both photosystems and produces NADPH, ATP, and oxygen:



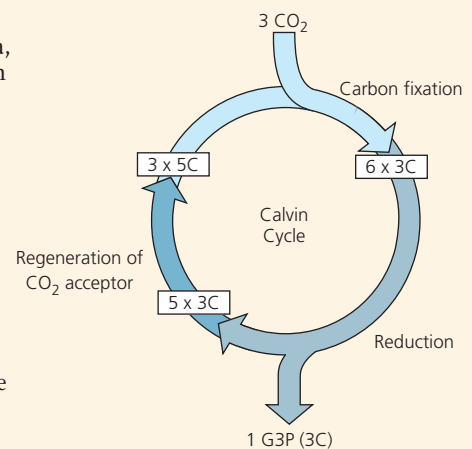
- **Cyclic electron flow** employs only one photosystem, producing ATP but no NADPH or O<sub>2</sub>.
- During chemiosmosis in both mitochondria and chloroplasts, electron transport chains generate an H<sup>+</sup> gradient across a membrane. ATP synthase uses this proton-motive force to make ATP.

? The absorption spectrum of chlorophyll *a* differs from the action spectrum of photosynthesis. Explain this observation.

### CONCEPT 10.4

#### The Calvin cycle uses the chemical energy of ATP and NADPH to reduce CO<sub>2</sub> to sugar (pp. 201–202)

- The Calvin cycle occurs in the stroma, using electrons from NADPH and energy from ATP.



- One molecule of **G3P** exits the cycle per three CO<sub>2</sub> molecules fixed and is converted to glucose and other organic molecules.

**DRAW IT** On the diagram above, draw where ATP and NADPH are used and where rubisco functions. Describe these steps.

### CONCEPT 10.5

#### Alternative mechanisms of carbon fixation have evolved in hot, arid climates (pp. 203–206)

- On dry, hot days, **C<sub>3</sub> plants** close their stomata, conserving water but keeping CO<sub>2</sub> out and O<sub>2</sub> in. Under these conditions, **photorespiration** can occur: **Rubisco** binds O<sub>2</sub> instead of CO<sub>2</sub>, consuming ATP and releasing CO<sub>2</sub> without producing ATP or carbohydrate. Photorespiration may be an evolutionary relic, and it may play a photoprotective role.
- C<sub>4</sub> and CAM plants are adopted to hot, dry climates.